User Manual

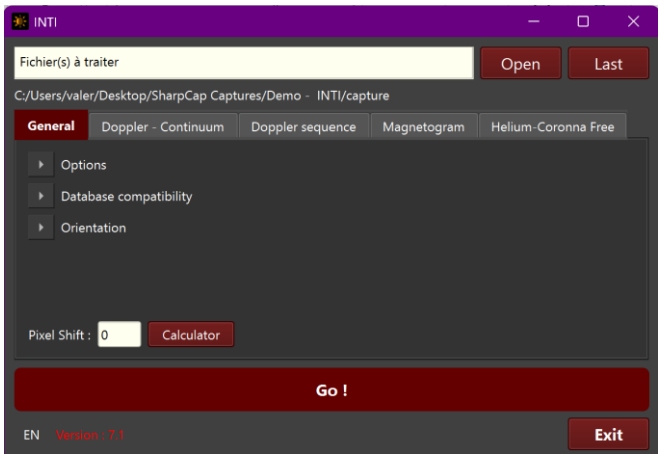
INTI V7

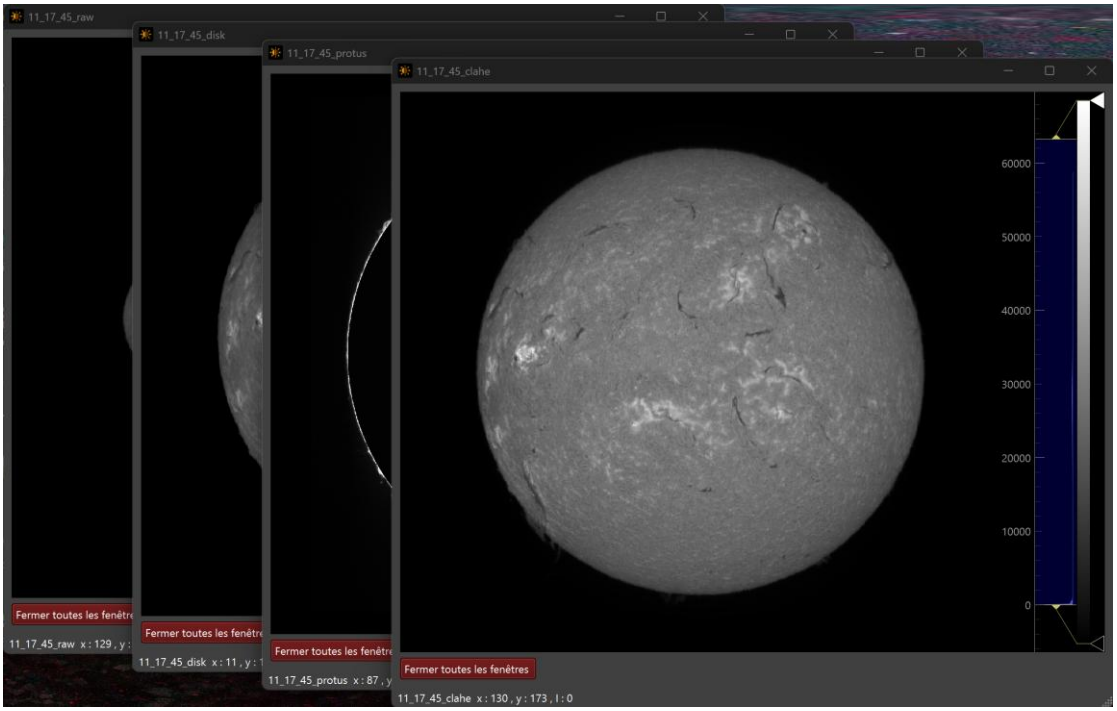
INTI is the first amateur application to automatically generate images from spectroheliograph scan files. It is the native application for the Sol'Ex spectrograph, but it can also be used with other SHGs whose geometry is based on the Sol'Ex design. Particular attention has been paid to the quality of the images produced while minimizing excessive processing. Its streamlined interface provides the essential functions an observer needs during an acquisition session. For clarity, post-processing and analysis functions are available in a separate software package: INTI Partner.

INTI processes monochrome SER video files in 8-bit or 16-bit format.

The INTI experience

Load one or more SER files, click Go!, and get your images.





Installation

1. Download the ZIP file

For Windows: http://valerie.desnoux.free.fr/inti/inti-window.ziphttp://valerie.desnoux.free.fr/inti/inti-window.zip

For Mac: http://valerie.desnoux.free.fr/inti/inti_macos.zip. Refer to the appendix of this document for instructions on handling the security permissions required for software not certified by Apple.http://valerie.desnoux.free.fr/inti/inti_macos.zip

For Linux: http://valerie.desnoux.free.fr/inti/inti-linux.zip. As Linux distributions vary, the provided build targets the latest version of Ubuntu.http://valerie.desnoux.free.fr/inti/inti-linux.zip

2. Extract the INTI folder to your hard drive or a USB drive.

Once extracted, you will have a folder containing an _internal folder with a number of files, as well as the inti.exe executable.

3. Click INTI.exe to launch the application

The executable file is located in the extracted folder.

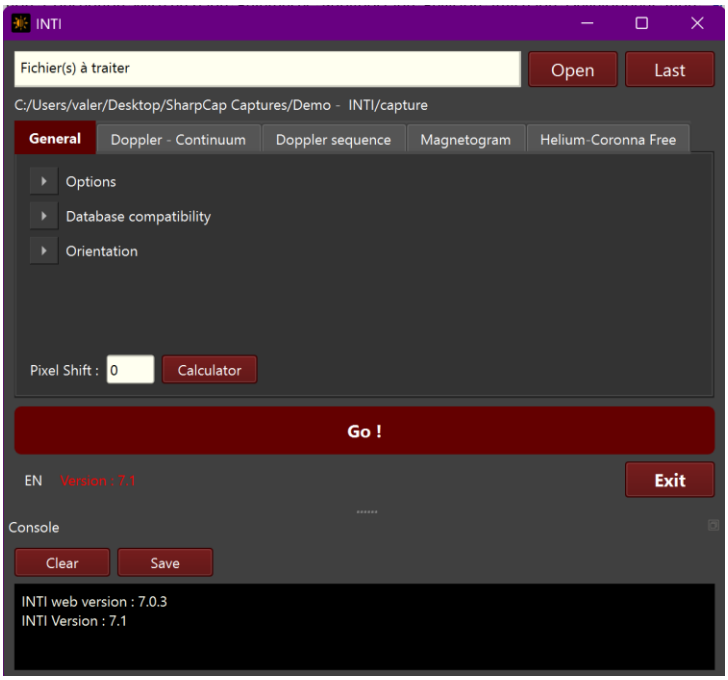
The user interface may take ten seconds or more to appear, particularly on macOS. It will launch faster the second time.

When launched, Windows may not recognize the application's digital signature and may display a warning to protect your PC against viruses, preventing the program from running. If you trust the program, choose the option to run it anyway.

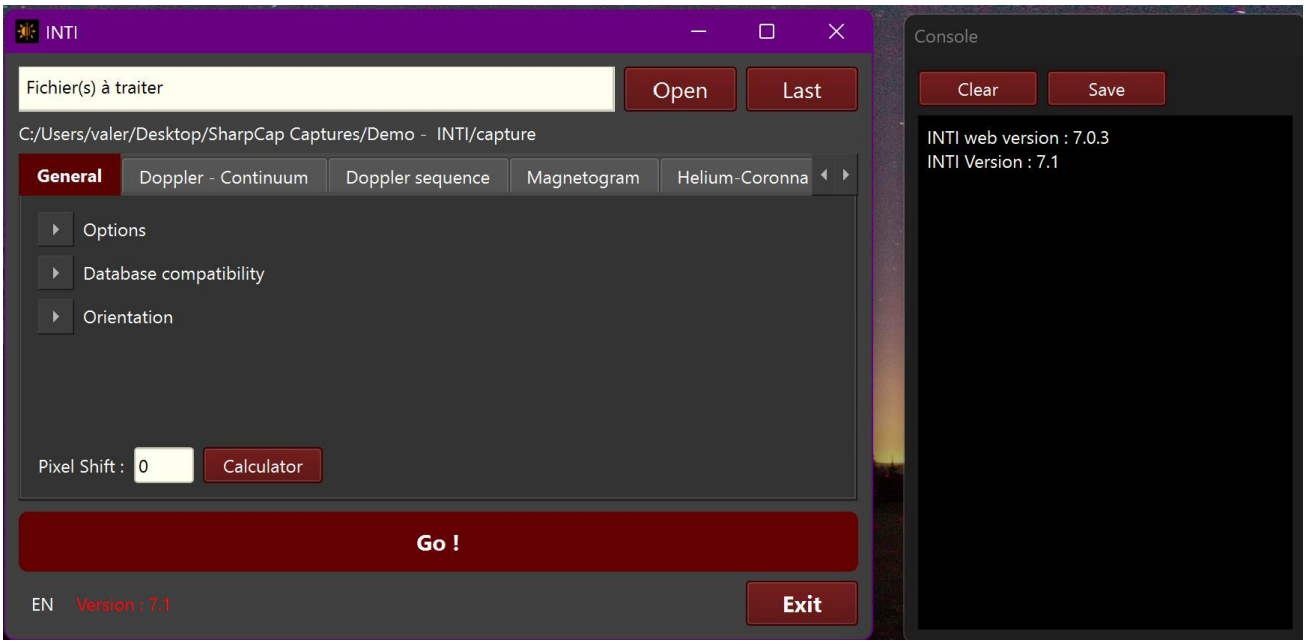
If you encounter problems with the inti.yaml file, delete it and restart the program. Make sure that the folder has write permissions.

Overview

When launched, the window shown below appears. It consists of a tab area and a panel at the bottom (dock) called Console, which displays information throughout processing.



The application window can be enlarged. The Console can independently be positioned on the left or at the bottom, or detached as a separate window. To do this, click near its title and the three dots, then drag while holding down the mouse button. To dock the floating Console again, double-click its title bar.



The application remembers your interface layout for the next launch.

Language settings

To change the default language from French to English, click the ‘FR’ button and then restart the application.

Version check

If you have an Internet connection, the application checks the current version number on the website. If it differs from the installed version, the version indicator turns red.

Tabs

Each tab corresponds to a way of processing the SER file. This makes it easy to view the settings and access the relevant processing function directly.

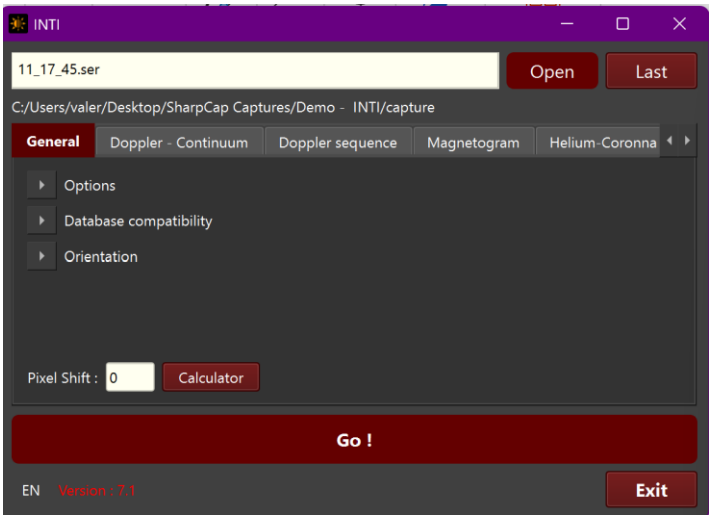
The first tab, “General”, provides a number of options and settings for customizing your preferences.

General processing

Click “Open” to browse to the folder containing the file or files. You can also press the space bar to open the file selection dialog.

Select the file(s).

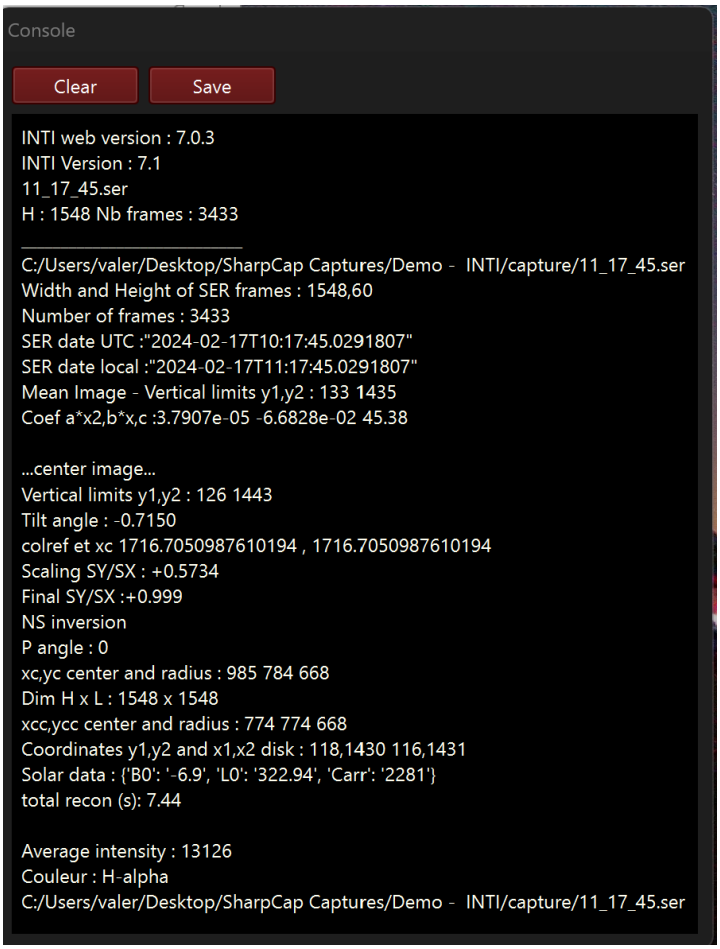
Click “Go!” to start processing. You can also press Enter/Return to launch processing.



The folder name is displayed below the file-name field.

If a folder has been specified, you can process the most recent SER file in that folder by clicking “Last”, without having to select a file. This is particularly useful for checking acquisition quality during an observing session.

Information is displayed in the Console throughout processing. It is also saved in a file named “SER_filename_log.txt” in the same folder as your SER file. You can clear the Console with the “Clear” button and save all its text to console.txt with the “Save” button.



INTI generates the output image by locating the darkest spectral line in your acquisition, using an image formed by averaging the frames. INTI then calculates the shape of the slit projected onto the image and uses this model to extract, frame by frame, the intensities at the core of this line. The “raw” image is then processed to produce a circular image of the solar disk:

- Line correction, to remove defects caused by impurities on the slit
- Non-uniformity correction
- Geometric correction using tilt correction (when the scanning direction is not perpendicular to the slit)
- Ellipticity correction (when the solar disk drifts too slowly or too quickly relative to the acquisition frame rate).

It is possible to generate an image at a wavelength shifted by a few pixels from the spectral-line core. To do this, enter a value in the “pixel shift” field.

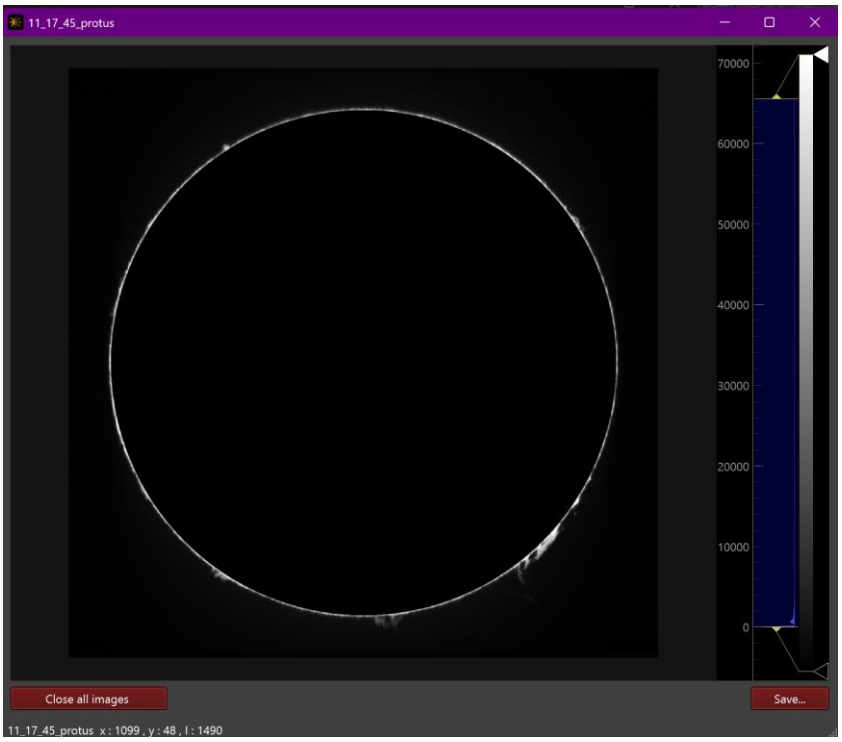
Once processing is complete, the resulting image files are displayed.

- **Raw: the uncorrected disk, with no cross-slit or geometric corrections. This is provided for a quick check after acquisition, so that you can adjust the exposure time or frame rate to obtain an almost circular disk.**
- **Disk: the final image with predefined contrast levels and no post-processing, or Cont if a wavelength shift was requested.**
- **Protus: the final image with levels adjusted to show faint prominences, using a virtual occulting disk.**

- **CLAHE: the final image with “Contrast Limited Adaptive Histogram Equalization” post-processing applied.**

Image window(s)

For each image:



When the mouse pointer is over the image, use the scroll wheel to zoom in or out. Hold down the left mouse button to move the image within the frame.

Moving the mouse over the image displays the intensity of the pixel under the pointer in the x, y, Intensity readout at the bottom of the window.

Right-clicking the image opens a context menu. Select “View All” to return to a centered view of the entire image, or “Export” to export it as a PNG file, in addition to the PNG image already saved by the application.

You can change the display levels by moving the histogram sliders on the right with the mouse. You can also save the modified image using the “Save…” button. A log.txt file will be added with the same base name as the new image, for use in INTI Partner.

If the pointer is over the histogram, the mouse wheel can be used to zoom in on the histogram as needed.

The “Close all windows” button closes all Image windows displayed after processing an SER file. You can also use the “Esc” keyboard shortcut.

When processing a batch of files, the images are updated at the end of each file’s processing. You can track progress in the title bar of the last window, in the following format: processed filename… next next filename (n / number of files to process).

Quitting the application

Click the “Exit” button to close the application and save your default choices in the inti_ini.yaml file.

Output images and folders

All results are saved in the folder containing your SER file. Processed file names always begin with “_”, followed by the SER file name without its extension.

INTI creates three subfolders in your working directory:

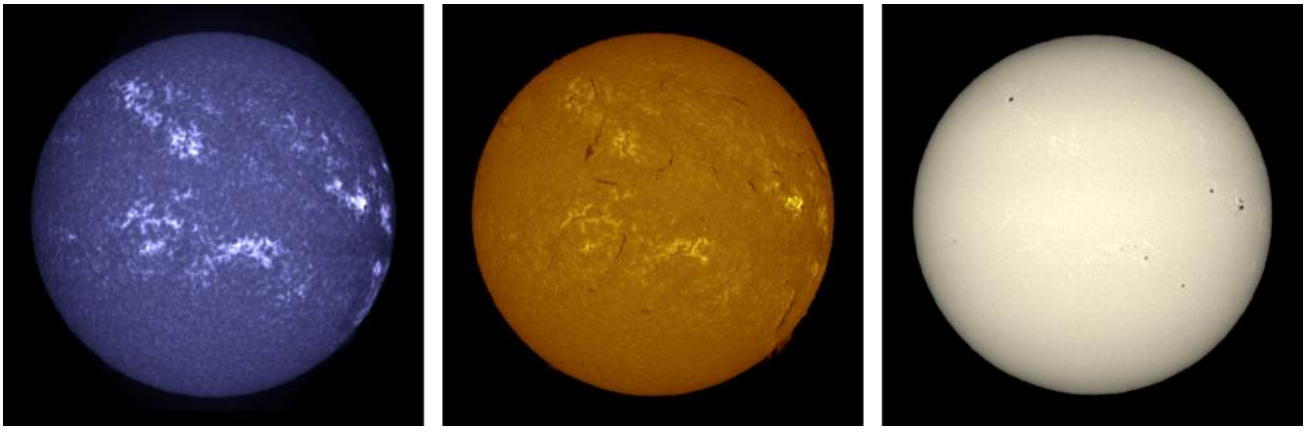
- **BASS2000: stores the three files generated for submission to BASS2000: two FITS files (disk and protus) and a JPEG thumbnail. This folder is created only if the wavelength is selected in the “Database compatibility” panel. See the “Database compatibility” section in this document.**
- **Clahe: contains all PNG files processed with CLAHE, for quick inspection.**
- **Complements: contains raw and averaged files, as well as other complementary FITS and PNG images.**

PNG images

- _disk.png: a view of the final image with low contrast levels, preferably used for your own post-processing; or _cont.png if a shift value was entered.
- _protus.png: a view of the final image with a black occulting disk to highlight prominences.
- _clahe.png: the final image with “Contrast Limited Adaptive Histogram Equalization” applied, stored in the Clahe folder.

Depending on your preferences, additional PNG images may be generated.

- _color_xx.png: a colorized image using one of three colors automatically selected by INTI, with color profiles for Calcium, H-alpha, or Pale (for the continuum).
- _raw.png: a monochrome view of the solar disk as scanned, built line by line and stored in the Complements folder. Useful for analyzing the SER file in INTI Partner.
- _grid.png: an image with a Stonyhurst grid, stored in the BASS2000 folder if Stonyhurst is enabled in the advanced settings.
- _inv.png: an image with inverted display levels.
- _mix.png: a composite image combining prominences and the disk.



Example of automatic colorization by INTI, from left to right: Calcium image, H-alpha image, and continuum image.

FITS images

- _recon.fits: the final monochrome FITS image, referred to as the “reconstructed” image.
- _raw.fits: the raw image, stored in the Complements folder.
- _mean.fits: the spectral image obtained by averaging all frames, stored in the Complements folder.

Depending on your preferences, additional FITS images are saved in the Complements folder in advanced processing modes.

- _mean_start.fits: in free-line mode (Helium), the average spectral image around the time the spectrum first appears, used for automatic emission-line detection.
- _diff.fits, _sum.fits, _x0, x1, x2: intermediate FITS images used to build a magnetogram.
- Dop1, dop2, moy: intermediate monochrome FITS images used to build the Doppler image.

DAT file

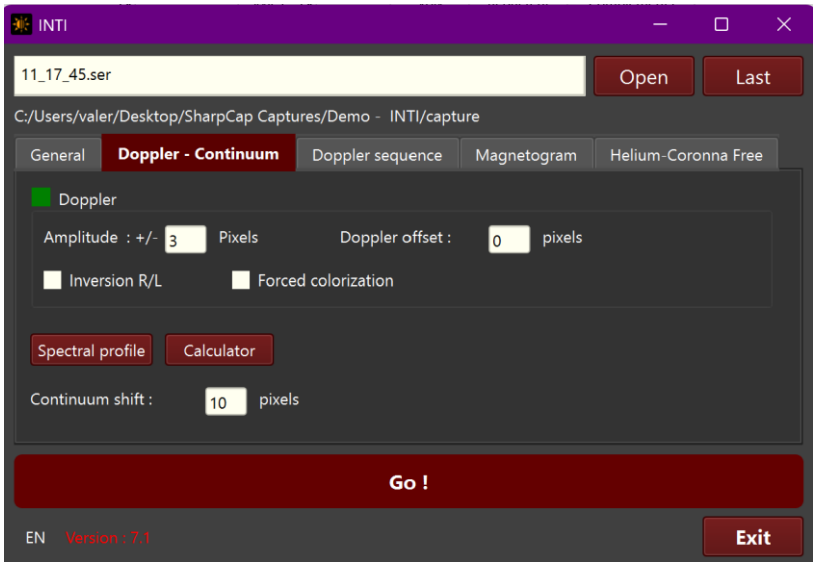
- .dat: the spectral profile of the _mean.fits image in the Complements folder. It can be viewed with Visual Spec, ISIS, Excel, etc.

Note: except for CLAHE images, no processing changes the intensity ratios between pixels. No gamma adjustment or sharpening is applied.

Doppler and Continuum

In addition to the line-core image, this reconstruction mode calculates a composite image called “Doppler” and an image shifted close to the spectral continuum, providing a conventional view of sunspots.

You can generate only the continuum image in addition to the line-core image by disabling the Doppler option.

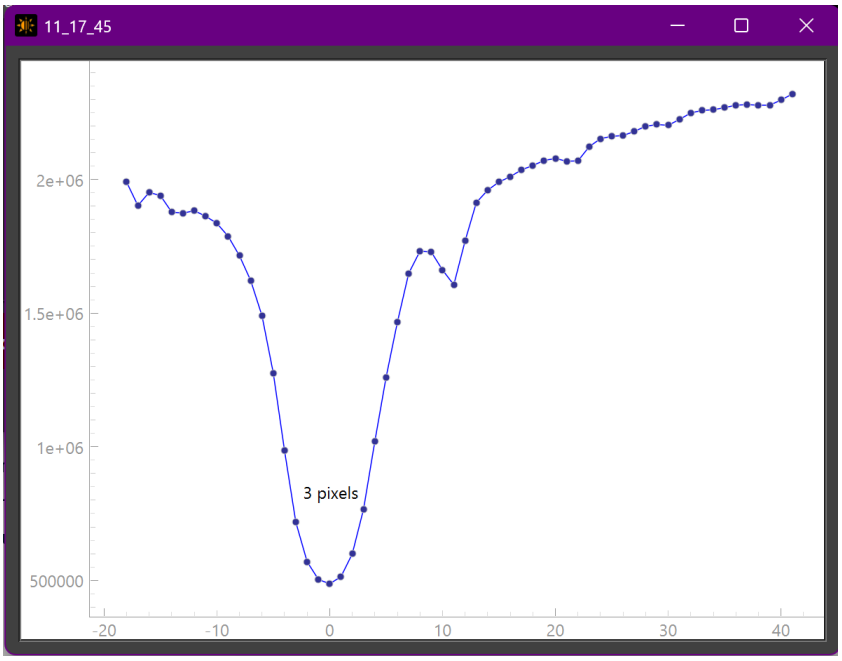


The Doppler image is composed of two images with the same wavelength offset on either side of the central wavelength. The value is set in the Amplitude field. The Doppler offset can be used to shift the Doppler values away from the center. The final color image is composed of the +n-pixel

image in the red channel, the -n-pixel image in the blue channel, and the average image (R+B / 2) in the green channel, with intensity normalization.

An amplitude of 3 pixels calculates images at -3 and +3 pixels relative to the line center. A Doppler offset of 1 pixel calculates images at -2 and +4 pixels.

To help you understand the line profile and the positions of the Doppler offsets, you can display the average spectral profile using the “Spectral Profile” button. Click one of the points on the profile to see its offset relative to the line center. If the SER file has never been processed, this profile file is not available. The .dat file is saved in the Complements folder.

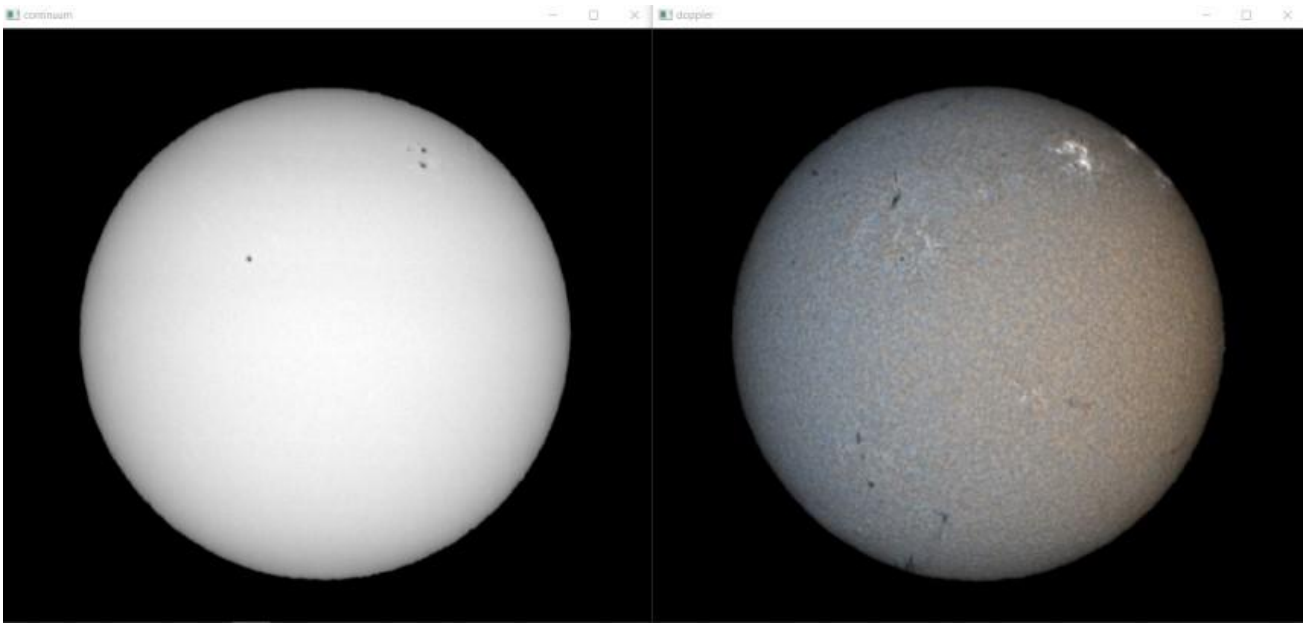


You can also reverse the order of the blue and red wings using the L-R inversion option, since this depends on the scanning direction during acquisition.

At the end of processing, if no error is detected, two PNG images will be displayed and saved: _doppler.png (color image) and _cont.png (continuum image).

The prominence image is also generated from the Doppler image, using levels adjusted to show the Doppler effect.

A high-intensity Doppler color mode is available to force red-red and blue-blue colors, or white when the velocities cancel out.



Left: continuum image; right: Doppler image.

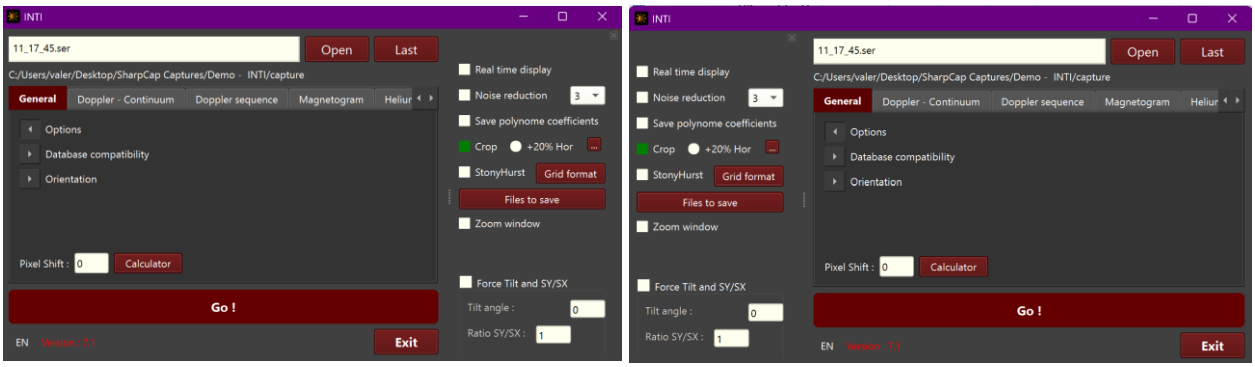
Three FITS images are also created:

- xx_recon_cont_continuum shift value.fits
- xx_recon_dp_-amplitude value.fits (image at -n pixels)
- xx_recon_dp_+amplitude value.fits (image at +n pixels)

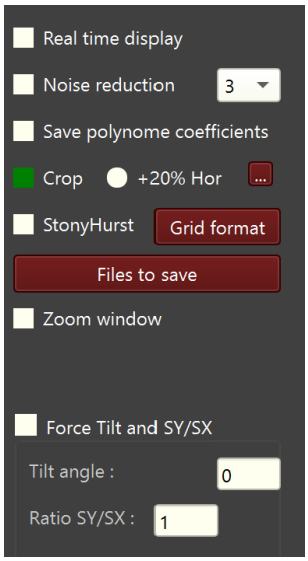
An error will be reported if the pixel shifts fall outside the image along the slit length.

Advanced options and preferences

When you click the Options section in the General tab, a panel appears on the right. Like the Console, this panel can be moved, but it cannot be detached from the application.



Interface layouts with the panel on the right or left. The panel is on the right by default.



Advanced options panel

Real-time display: INTI displays the disk reconstruction line by line as the scan is processed. This takes longer, but you can watch the disk being built in real time alongside the corresponding spectral video. This is very useful for demonstrating how a spectroheliograph works.

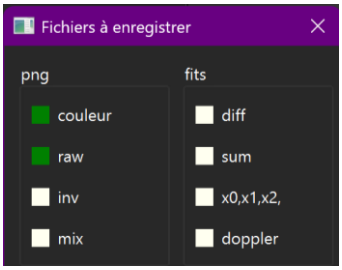
Noise reduction: averages 3 or 5 pixels around the line core, selected from the drop-down list. This reduces noise but also reduces spectral resolution.

Save polynomial parameters: in some special modes, such as Magnetogram or Free Line, a strong, deep line may be present next to the weaker line of interest, which may not be visible during acquisition in the region of interest. You can use the strong line to calculate the polynomial in General mode and transfer its parameters to the special modes. For security reasons, this option is not saved in the inti_ini.yaml file.

Crop: forces automatic image centering and cropping. An option allows the automatic calculation to add 20% to the horizontal dimension, for example if a prominence extends beyond the frame in normal crop mode. The small "…" button opens the manual crop settings for exceptional cases.

Stonyhurst: generates a so-called Stonyhurst grid overlaid on the solar image. You can customize its colors and display format using the "Grid Format" button. This can also be done later in INTI Partner, where annotations can be added.

"Files to save" button: preferences for saving optional PNG and FITS files. By default, inverted and composite PNG images are not always generated. FITS images can be useful in special modes such as Magnetogram, Free Line, or Doppler when fine-tuning the acquisition procedure.

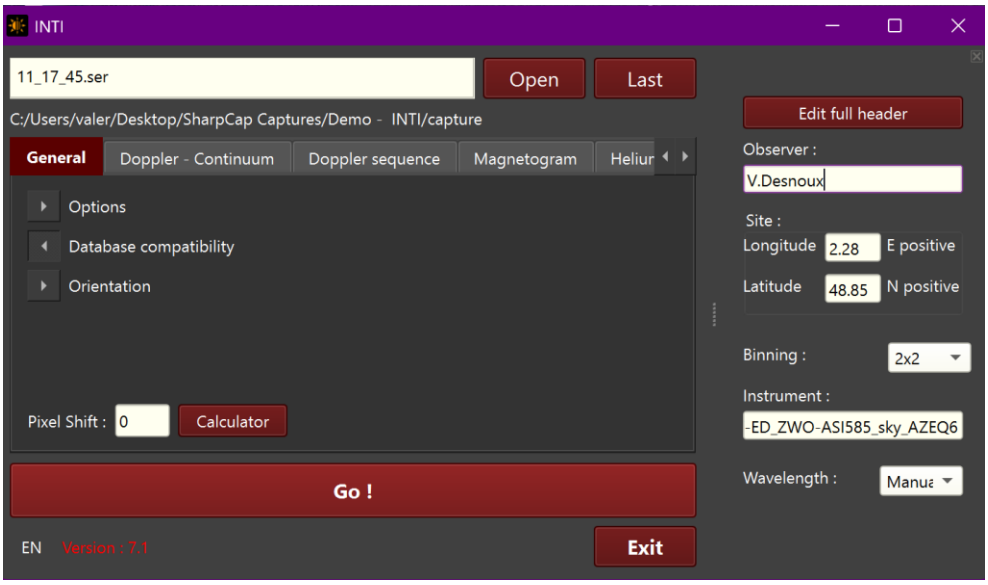


Zoom window: displays a 300 × 300-pixel mini-window showing the image at its native scale in the area under the mouse pointer, without having to use the mouse wheel or pan the image.

Force Tilt and SY/SX: forces a SY/SX scaling value and a tilt angle. The latest values are saved in inti_ini.yaml. If this option is not checked, these values are ignored and automatically calculated by INTI, then displayed in the Console.

Database compatibility

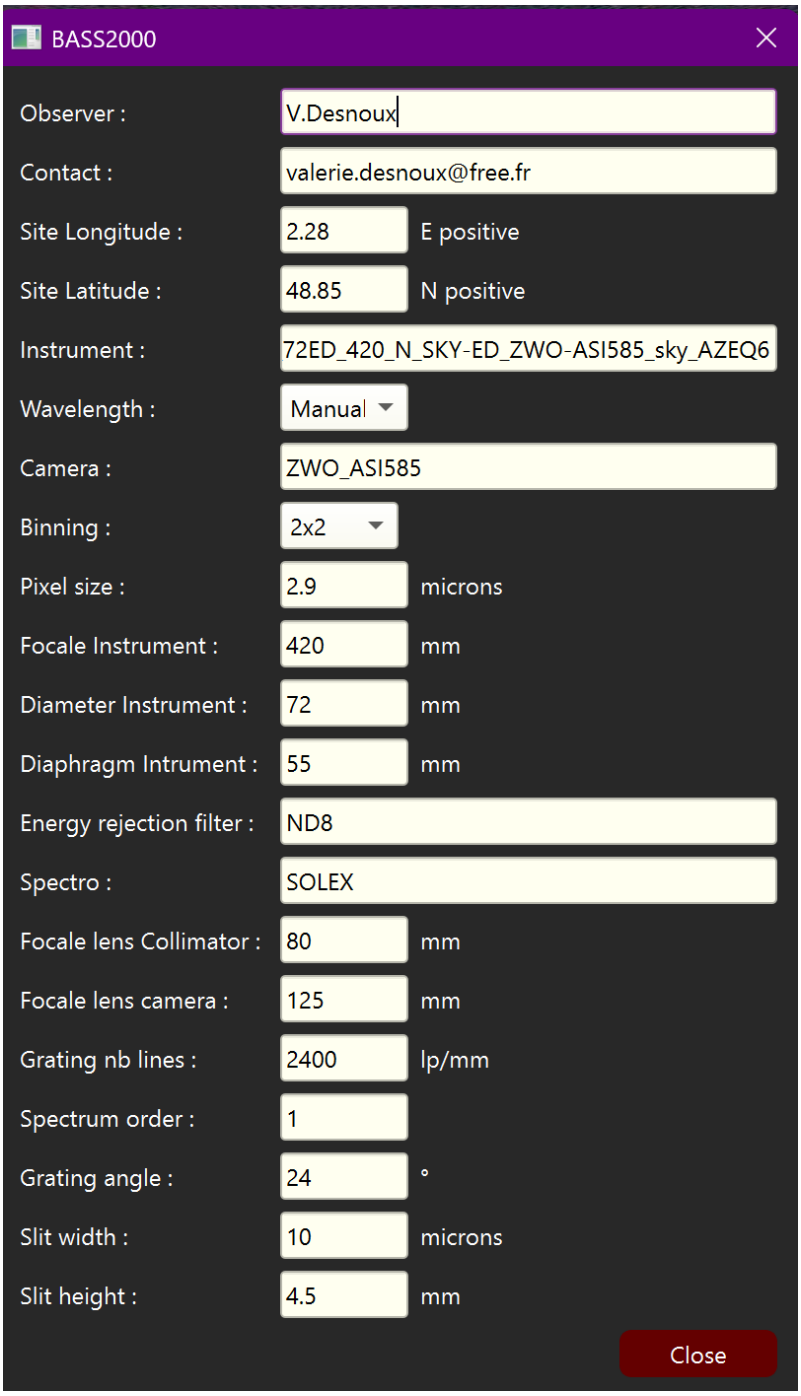
This panel allows you to add information to the FITS header to meet the submission requirements of the SOLEX database at BASS2000, a professional systematic-observation database managed by a team at the Paris Observatory in Meudon, known as the SOLAP database.[BASS2000]



The information directly accessible in this panel is that which may change between observing sessions: observing site, binning value, instrument name, and observed wavelength.

Other parameters are also required and can be accessed using the “Edit Full Header” button. However, they do not need to be changed unless you change your spectrograph or optical instrument.

All these parameters are saved in a “bass_entete.json” file, which can be read and edited outside the application. It can also be transferred from one PC to another to restore your observing configuration.



L'ensemble des paramètres sont présentés, incluant les paramètres du panneau précédent afin d'assurer la bonne cohérence des informations. Se référer au manuel opérateur BASS2000 (français - anglais)

Nom de l'observateur, contact mail, coordonnées du site et nom de l'instrument.

Sélection de la raie spectrale observée - la liste est conforme aux spécifications de BASS2000. Si la sélection reste « manuelle » alors aucune image BASS2000 ne sera générée. Néanmoins, la sélection perdure si vous traitez une séquence dans une session mais ne se conserve pas si vous relancer l'application afin de limiter les erreurs de labélisation.

Valeur du pixel et binning de votre acquisition.

Diamètre et focale de l'instrument optique, diamètre du diaphragme si vous en avez placé un. Sinon, remettre la valeur de l'ouverture sans diaphragme.

Dispositif de réjection d'énergie, filtre neutre ou hélioscope.

Modèle de Spectrohéliographe si validé par Meudon.

Focale de la lentille du collimateur, de la lentille de la caméra, angle du réseau, nombre de traits, ordre du spectre

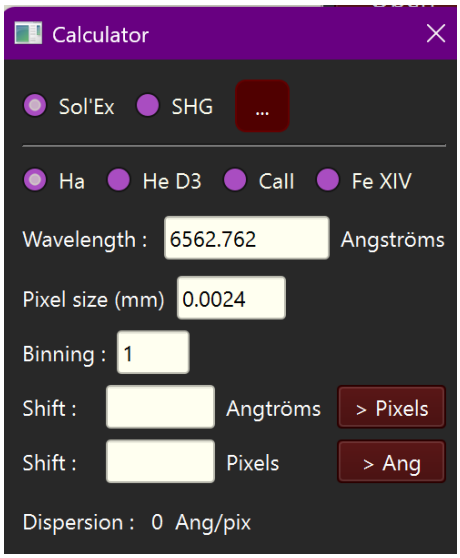
Fente, longueur et largeur

For submission, it is also essential to orient the solar disk correctly according to the conventions used by the solar community. The Orientation panel helps you do this.

Calculator

To check and convert the pixel shift into angstroms for the requested continuum images, use the calculator available from the "Calculator" button in the General tab. Note that the dispersion functions are specific to the optical geometry of Sol'Ex or an equivalent SHG that follows the original optical configuration.

Select the spectral line, then enter the pixel size and binning value. Enter either a value in angstroms and click “>pixels” to convert it to pixels, or a value in pixels and click “> ang” to convert it to angstroms. These calculations are specific to Sol’Ex optical geometry.

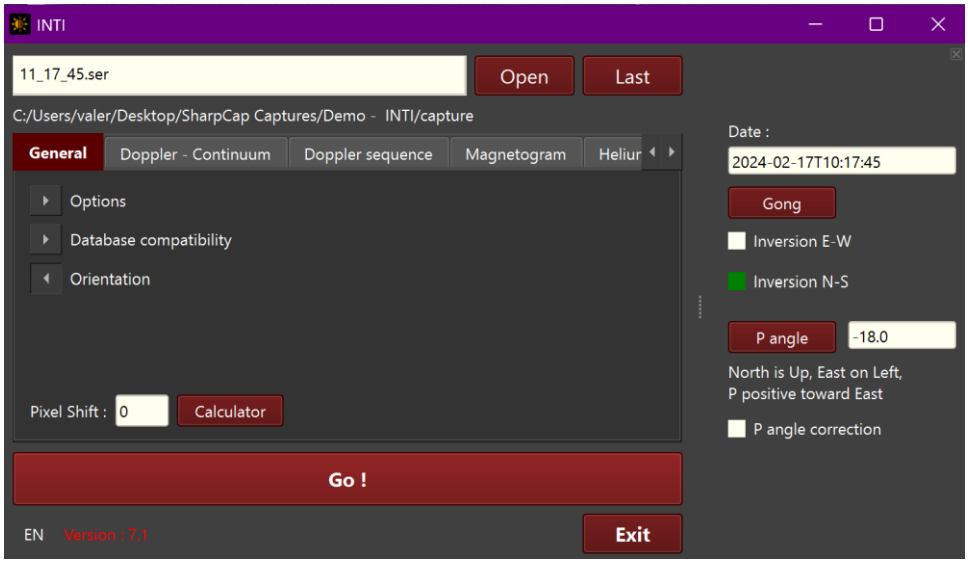


You can define the parameters of an SHG that differs slightly from Sol’Ex using the “…” button.

Only the camera lens parameters and grating angle determine the dispersion values.

Orientation

This panel provides assistance with correctly orienting the solar disk.



The solar-disk orientation convention is: solar (heliographic) north pole at the top, east on the left. The solar north pole differs from geographic north by an angle called the P angle. INTI calculates this angle internally from the UTC date stored in the SER file header.

By default, INTI does not apply the P angle to generated images. However, you can apply this angular correction systematically by enabling the Auto P Angle option. IMPORTANT: the orientation will be correct ONLY if the north-south and east-west inversions are correct. This correction is not applied to partial disks or to Magnetogram images, or Helium images with transversallium correction.

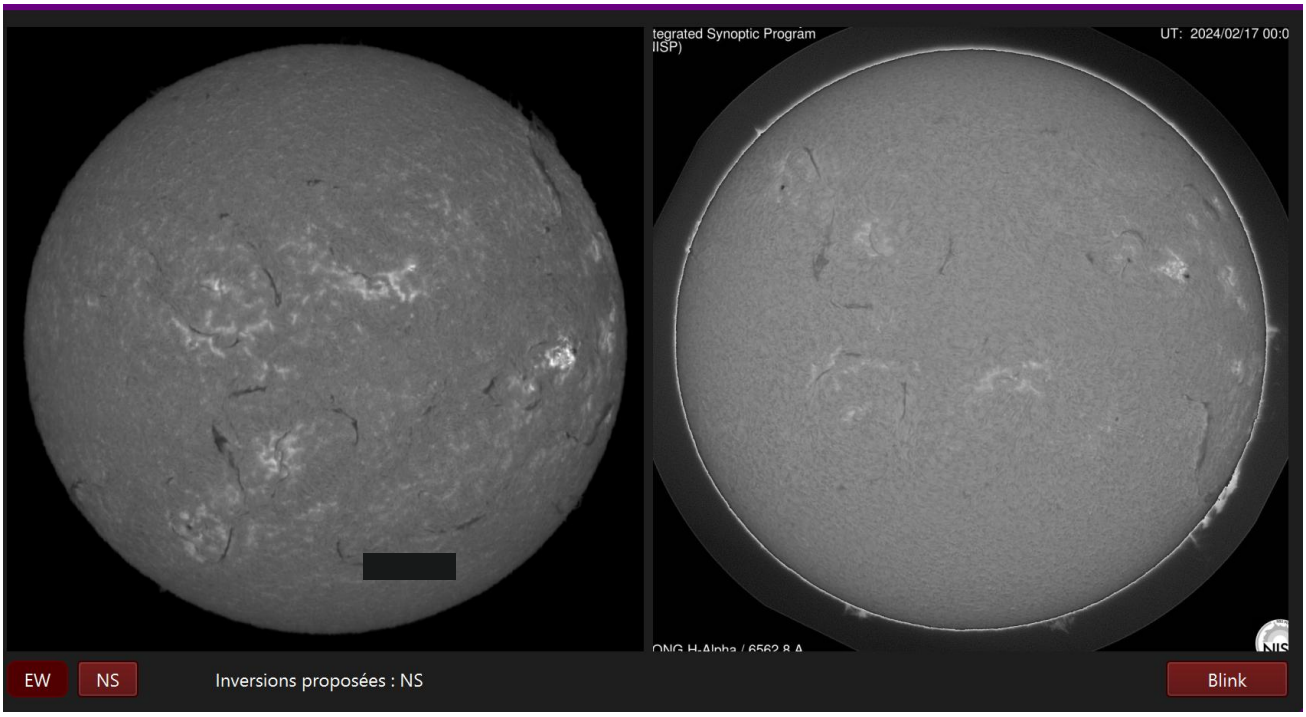
To determine the NS and EW inversions, you can compare your image with the closest image from the GONG network in date and time. (An Internet connection is required.) Click the “GONG” button. A comparison window appears, showing the _disk.png image on the left and the GONG network image on the right.

Click the EW and NS inversion buttons to find the correct orientation. The inversions are automatically transferred to the main interface when you close the comparison window.

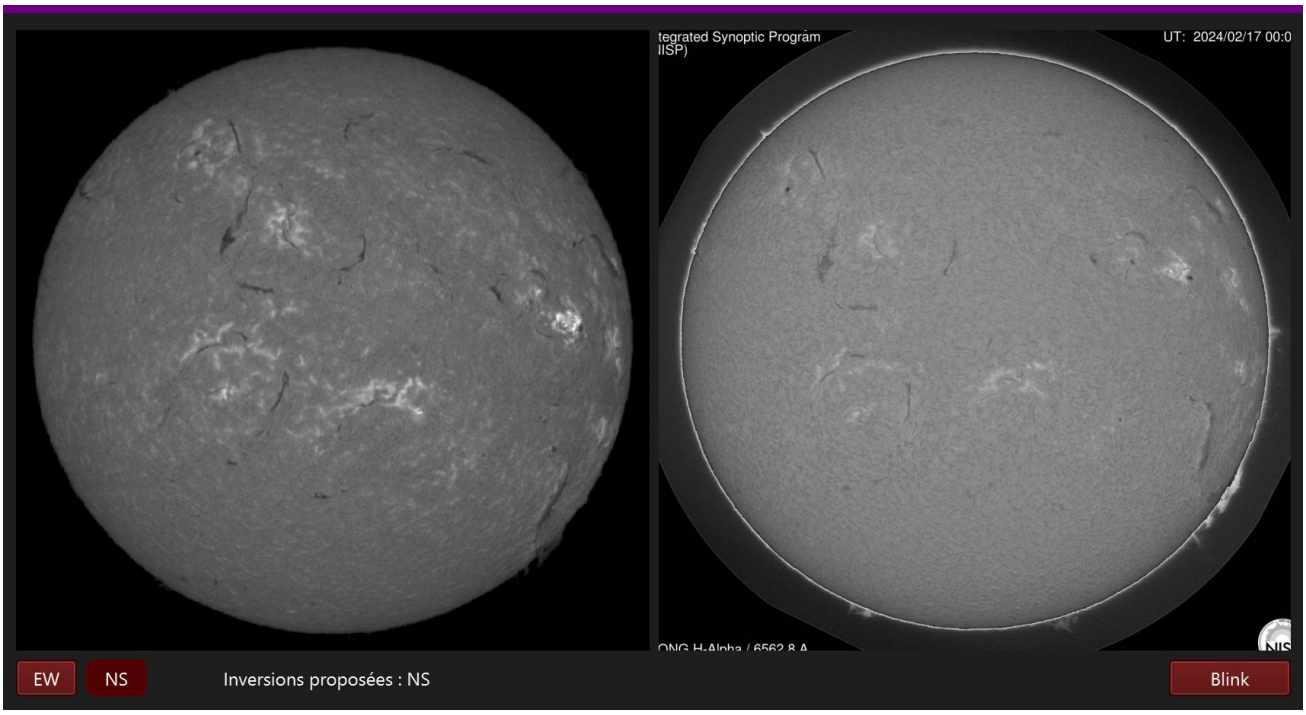
The “Blink” button alternates between the GONG image and your image in the right-hand pane.

You can resize the window with the mouse. The size is saved.

Note: an algorithm compares the two images and suggests the required inversions. Some GONG network images have low contrast, so this suggestion must be checked. It applies only to H-alpha images.



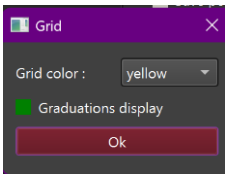
Comparison of the processed image with the GONG image. The suggested inversion is NS — click the NS button to check it.

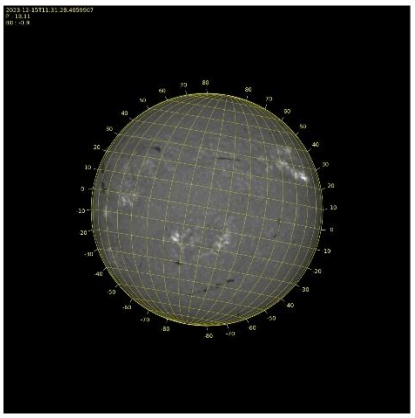


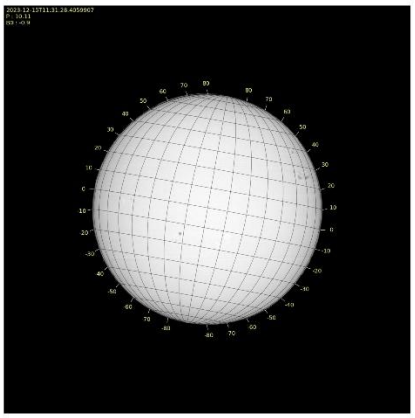
The processed image is oriented according to the suggestion. Check that it matches the GONG image.

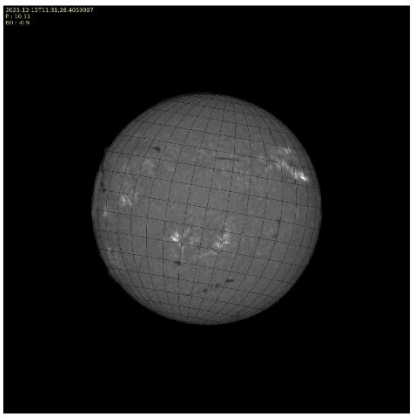
Stonyhurst grid

If the Grid option is enabled in the Options panel, an xx_grid.png file is saved in the BASS2000 folder. The grid format is defined using the “Grid Format” button, allowing you to adapt colors and display settings to the images. This grid is used to locate and track active-region centers. Note that you can recalculate the grid and add annotations later with INTI Partner, after your acquisitions.







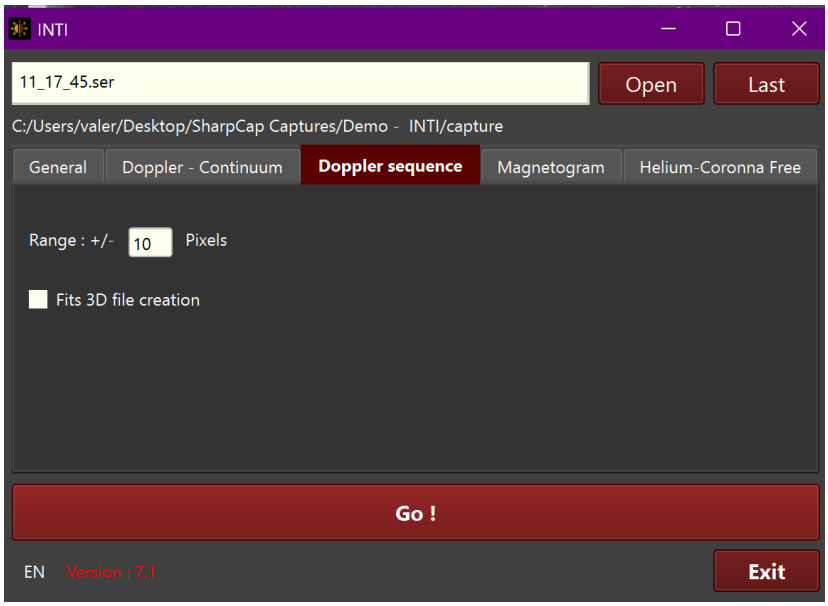


Doppler sequence

This mode automatically calculates an image cube in which each image is shifted by one pixel across a range of +/- n pixels.

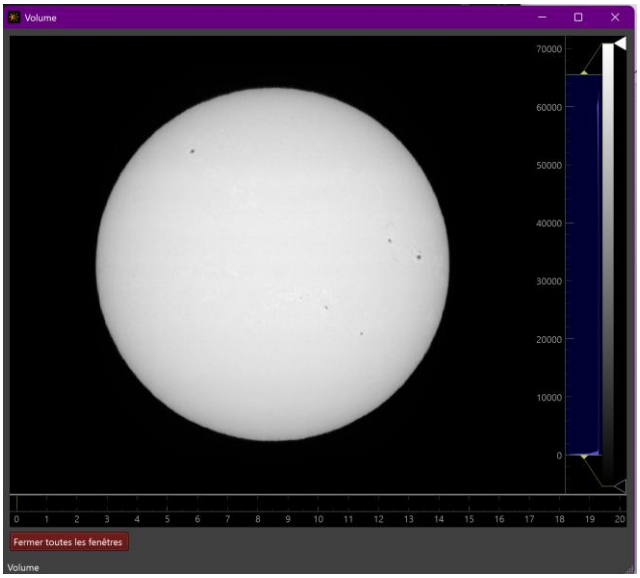
The result is saved as an animated GIF file and an MP4 file.

You can also enable saving in 3D FITS format.



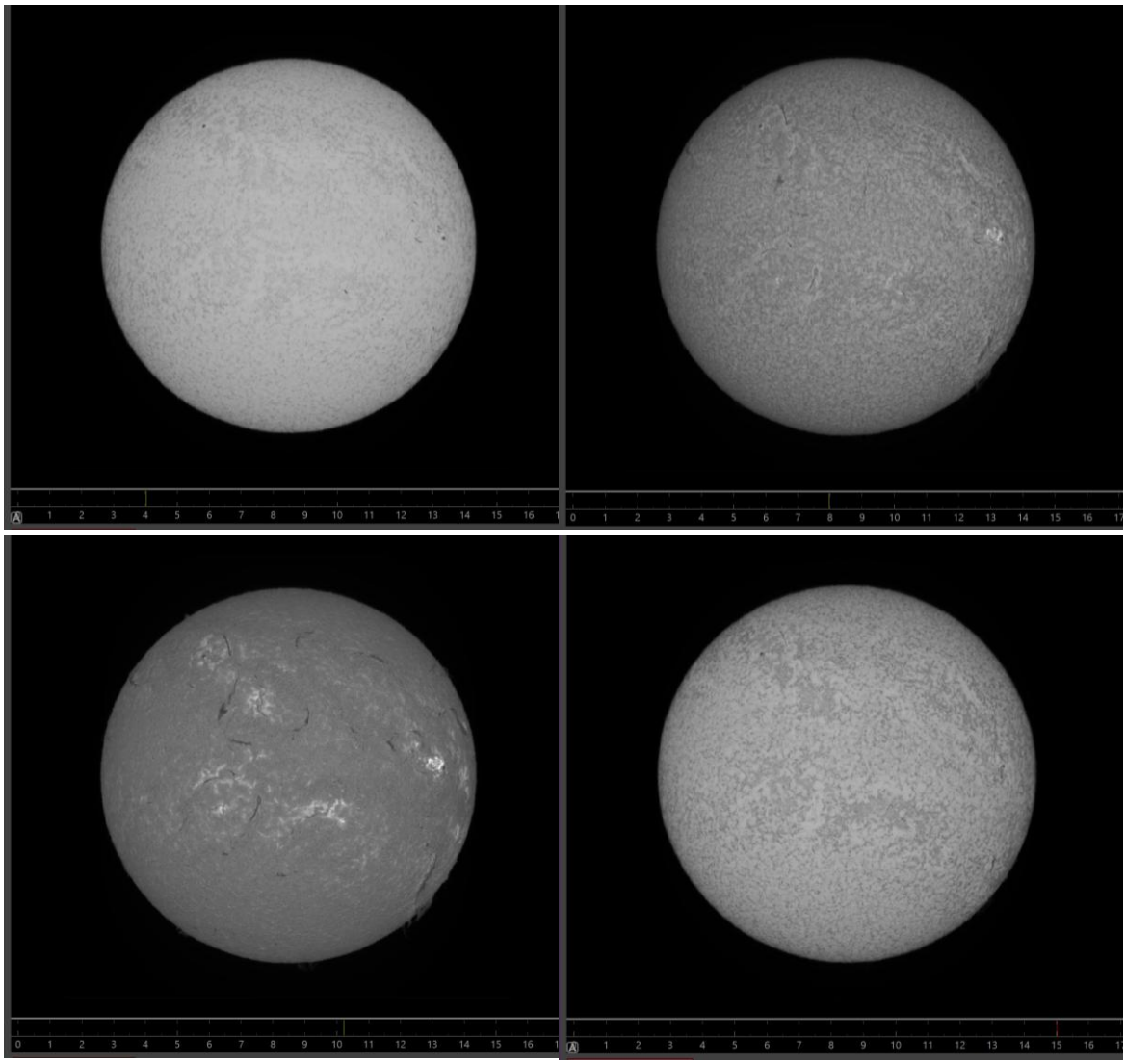
Select a file and specify the scan range as +/- n pixels.

Click Go!



The Volume window displays the image cube. Use the mouse to move the slider below the image and browse through the file frame by frame.

The GIF and MP4 files are saved in your observing folder.



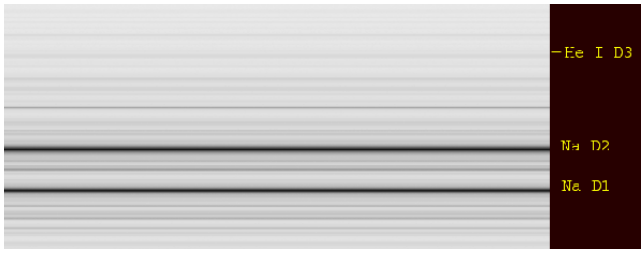
Frames 4, 8, 10 (central), and 15, corresponding to -6, -2, 0, and +5 pixels, converted to angstroms: -0.75, -0.25, 0, and +0.63 Å.

Helium line – Corona – Free-line processing

You can also refer to the observing guides available on the website.

A “free line” is a low-intensity line, weaker than one or more of the lines in the acquired spectral region. You must therefore locate this faint line and tell INTI where it is. It may even be invisible if you are looking for a line that appears only in emission at the solar limb. This is typical of Helium and solar-corona images.

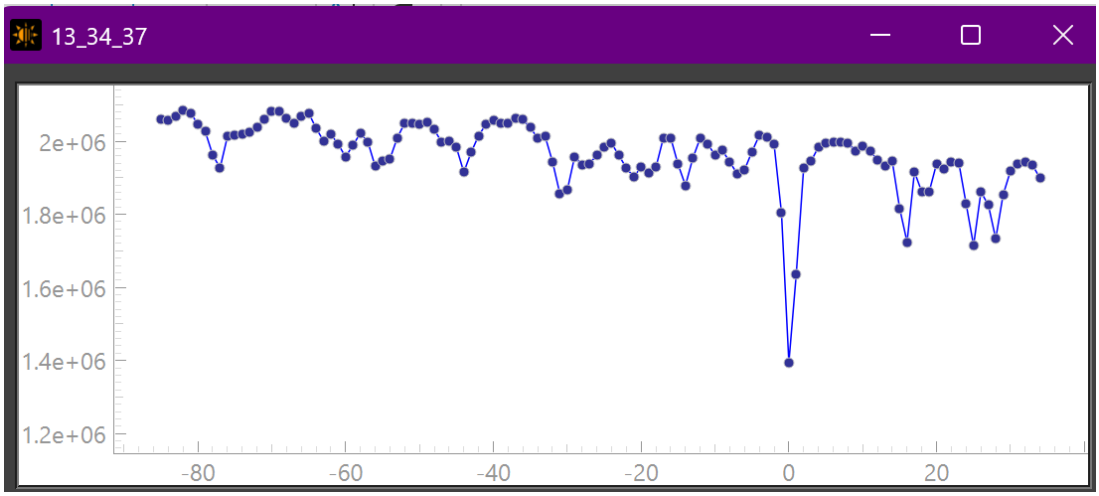
Let us consider the case of producing a Helium image in the D3 line at 5875 angstroms (INTI Partner – Map image).



Here is an image of one frame from the acquired SER file (INTI Partner – SER image).



Note the presence of a dark spectral line, which allows INTI to perform its automatic calculation. The remaining task is to locate the Helium line, which is almost invisible on the disk but visible in emission at the limb.



Spectral profile of the disk using the Profile tool in the Doppler tab. The darkest line is an Iron line and will serve as the reference line for the automatic calculation. Note that this profile cannot be used to determine the position of the Helium line.

For Helium, INTI can automatically find the emission line without any additional information and provide interactive tools to confirm its position if necessary.

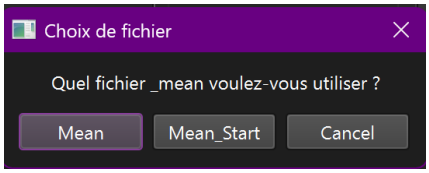
Polynomial coefficients

First, you need to perform an initial processing run to determine the polynomial coefficients (which model the curved shape of the slit image).

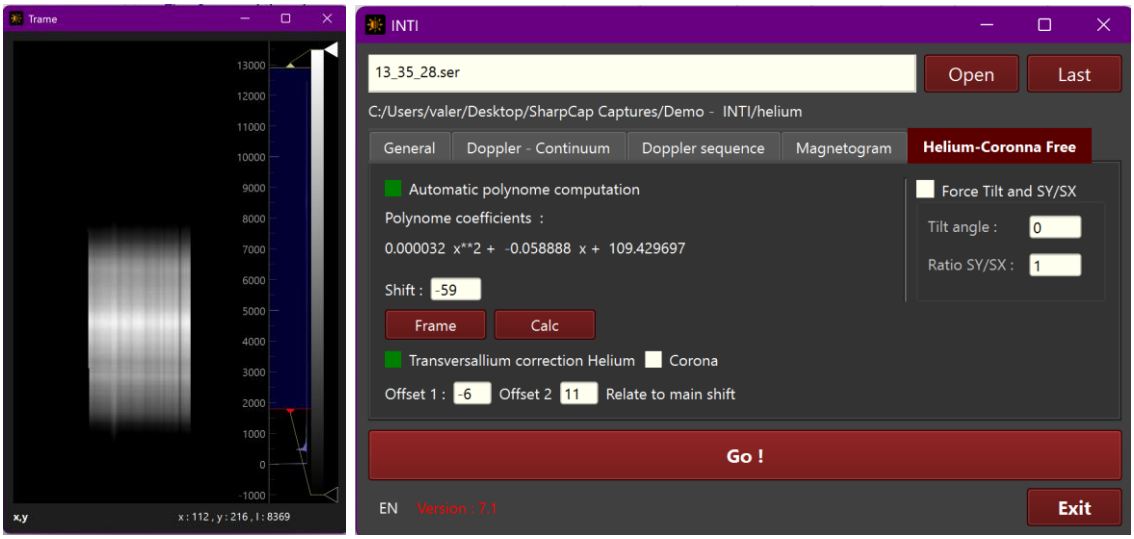
Enter your file name, click Automatic polynomial calculation, and then click Go!

Automatic location of the Helium line

Click the “Frame” button. A selection window appears. Instead of the average “mean” frame, it offers an average frame calculated automatically by INTI around the time the disk first appears: the “mean_start” frame.



INTI calculates a spectral profile and detects the position of the Helium line that appears in emission. The offset value is updated in the main interface.



Left: average spectral image at the time the disk first appears. The vertical white line that stands out and is also visible against the background is the Helium emission line. Right: the offset from the darkest line is updated in the interface. You can change the value by clicking the line position with the mouse and then clicking OK.

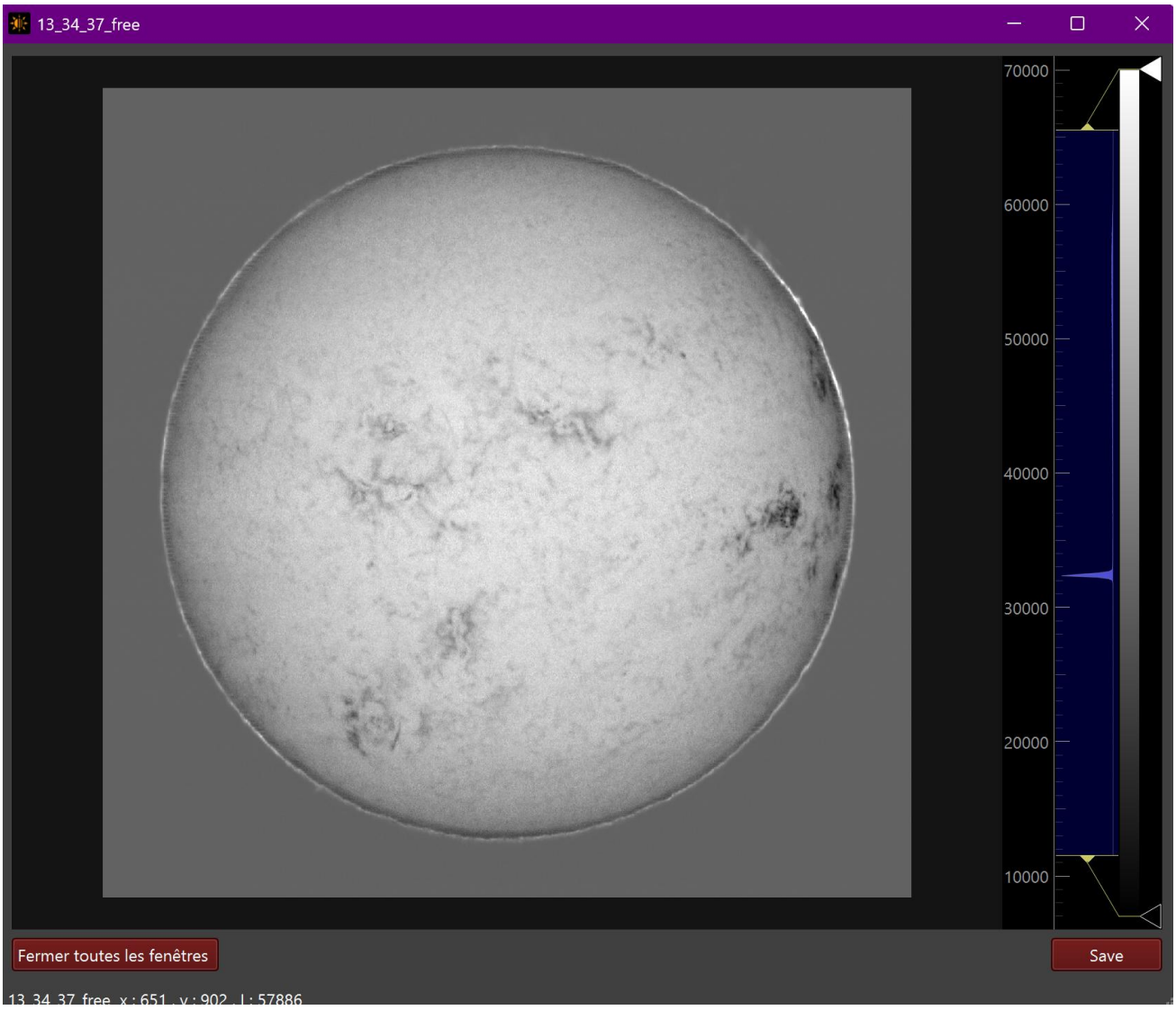
Important: the offset value is a constant determined by your camera (pixel size and binning factor). There is no need to recalculate it for every session or every scan. From now on, you can process all your Helium scans with a single click!

Note: for those who like doing the math, you can of course calculate the theoretical offset using the calculator available via the “Calculator” button. Pixel size is usually easy to find, but it can be harder to remember the binning mode used during acquisition.

Generating the Helium image

For a Helium image, you can add a horizontal-band correction, similar to a transversallium correction, by clicking “Helium transversallium correction”. Selecting this option automatically offers the “mean_start” image for detecting the Helium line.

Click Go!

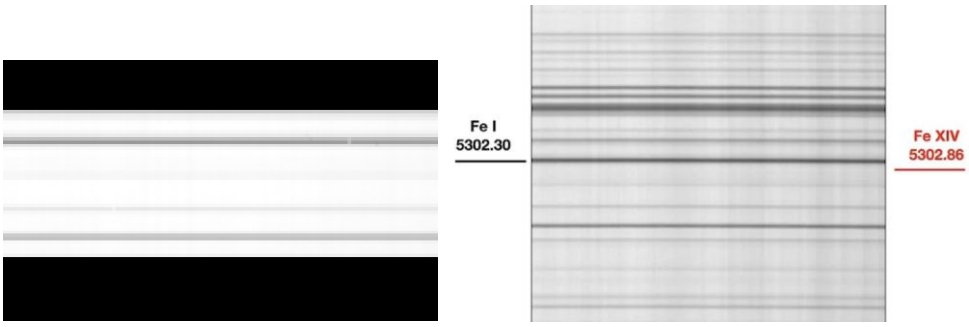


Solar corona image

The principle is the same: the coronal line is so faint that you need to use a nearby known line as a reference. Even if emission from the corona is detected, it is so faint that it cannot be seen directly in emission. A calculation is needed to determine the pixel offset corresponding to the wavelength difference, in angstroms, between the darkest line and the coronal line. In this example, the difference is 0.56 angstrom.

To improve signal quality, it is strongly recommended that you acquire data with the spectrum saturated.

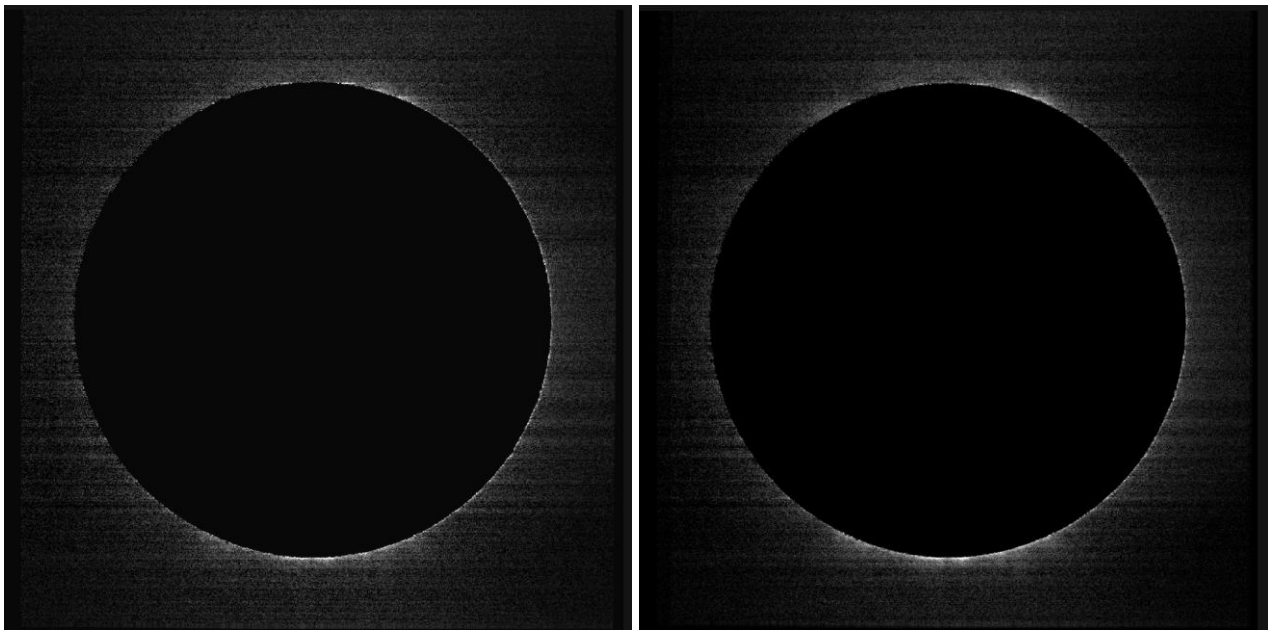
With the Corona mode in INTI, there is no longer any need to make an unsaturated scan to calculate the polynomial. INTI automatically tracks small line movements to obtain a better signal.



Left: spectrum from an unsaturated scan; right: line identification.

Choosing the continuum lines is another important step. You may accidentally specify an offset corresponding to a nearby weak line, as there are many lines in this region.

However, once the offset values have been defined, automatic polynomial calculation also works on saturated scans. It can track any slit movement to ensure that intensities are always extracted from the coronal line, which is both very faint and very narrow.



Left: coronal image obtained from a single scan; right: sum of two scans in INTI Partner.

Magnetogram

This reconstruction mode creates images of the solar magnetic field. The acquisition uses the Zeeman effect in certain iron lines that are particularly sensitive to magnetic fields. It requires polarizing filters and two scans. Refer to Christian Buil's video on the astro-spectro YouTube channel.**astro-spectro**

The processing procedure can be summarized as follows. The illustrations are provided by Olivier Aguerre.

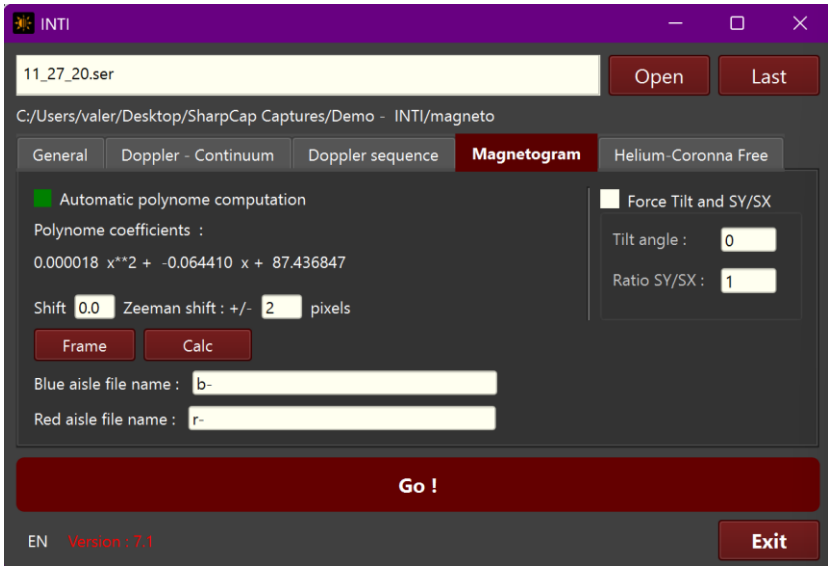
Processing the first scan – polarization A

Image of one frame from scan A (INTI Partner – SER image).

The selected Iron line is the line at 6302 Å (INTI Partner – Map tab).



The first step is to determine the offset between the darkest line and the Iron line at 6302 Å.



Select the file.

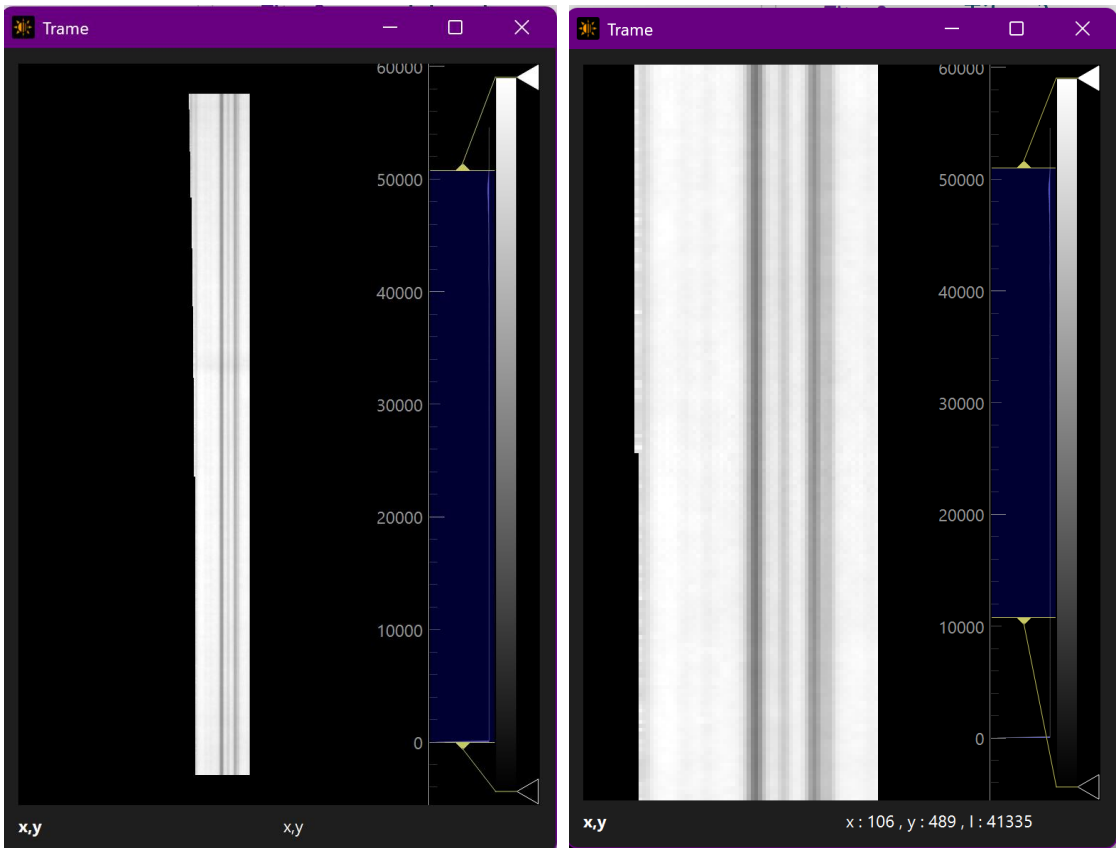
Click Automatic polynomial calculation.

Click Go!

The useful results are not the images, but the polynomial coefficients displayed at the end of processing.

To find the offset, click the “Frame” button.

An Image viewing window displays the spectral image (_mean), whose lines are straight because they have been rectified using the polynomial coefficients. You can zoom, pan, and adjust the display levels to focus on the two Iron lines.



Left: the _mean average frame, with straightened spectral lines; right: a zoomed view of the two Iron lines.

Move the mouse over the image to view pixel intensities. If you click the darkest line, the bottom of the window on the left displays: X: 89 (X position in the image), Offset: 0 (difference from the position of the darkest line; here it is 0 because you clicked that line).

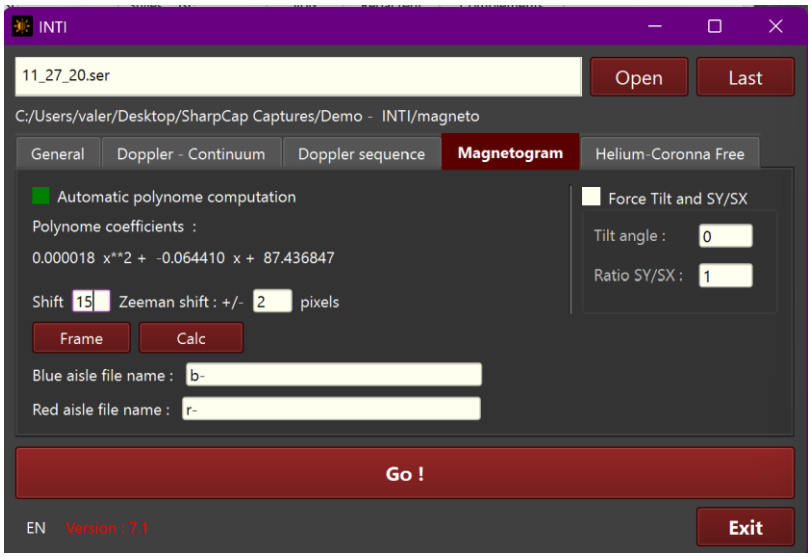


Continue moving the pointer until it is over the Iron line on the right, which is the line of interest. Then click its position.



The measured offset is 15 pixels. After you confirm by clicking OK, the main interface is updated with the value corresponding to the mouse click.

You can also calculate the theoretical offset using the calculator available via the “Calculator” button.



The hardest part is now complete. This offset is constant and specific to your camera (pixel size and binning). It can be calculated with the calculator, but finding it interactively is ultimately quicker and more reliable.

INTI calculates two images around this Iron line at 6302 angstroms, with offsets of +/- 2 pixels (you can of course change this value): one image in the blue wing of the line and one in the red wing. INTI prepares the file names. It is important to keep the hyphen at the end of the name.

Since you are processing the first scan, for polarization A, enter A in the name (or L for left, or Left; however, the same prefix must be used in both names).

Click Go!

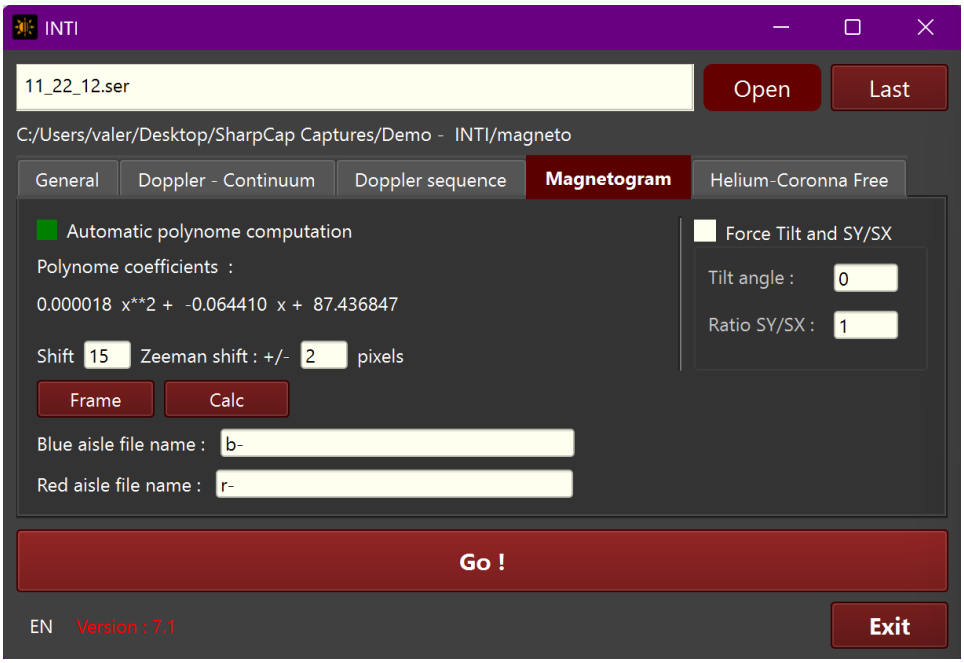
INTI saves the PNG and FITS images A_b-1 and A_r-1.

You can process a sequence of several files (batch processing); the images will be named A_b-1, A_b-2, and so on.

Processing the second scan - polarization B

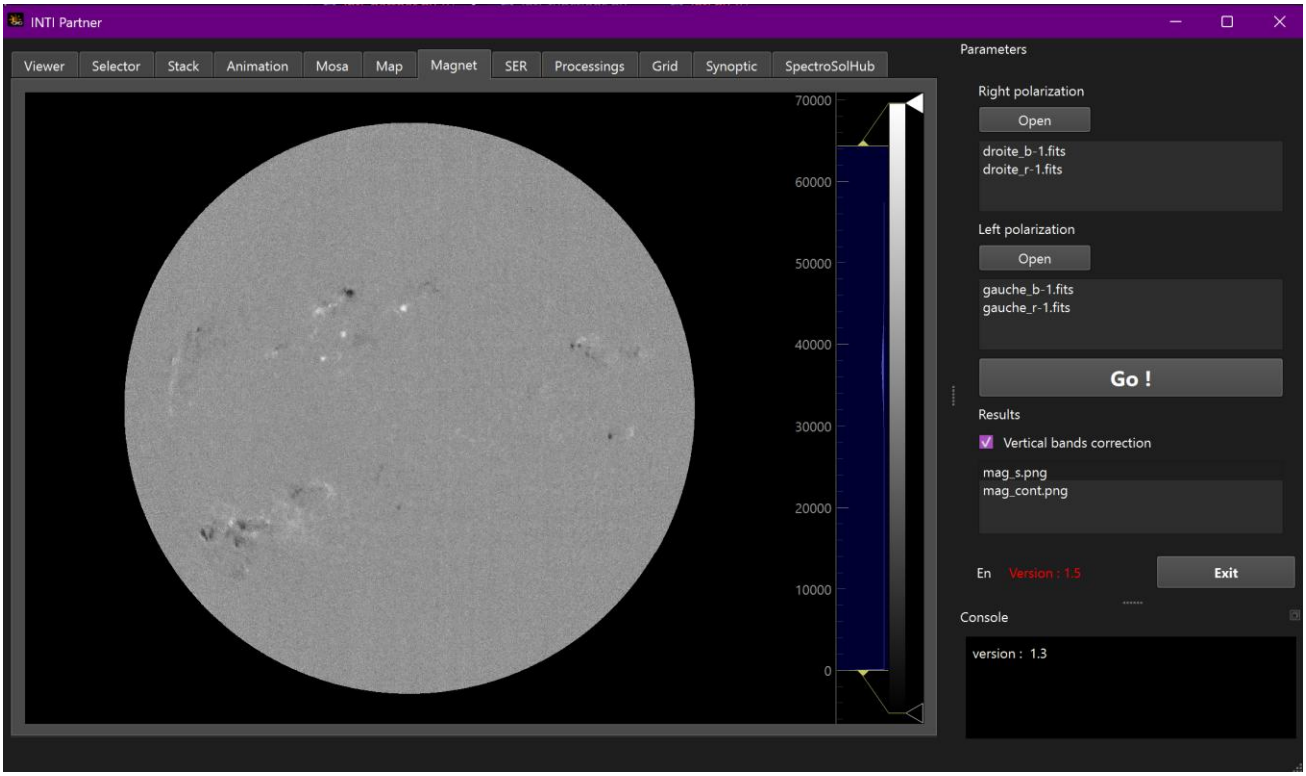
Simply enter the SER file name for the second scan (or the file names for batch processing), and change the blue and red file names to use the B_ prefix.

Simply click Go!



The working folder will now contain four files: A_b-1, A_r-1, B_b-1, and B_r-1.

To create the final magnetic-field image, open INTI Partner and run Christian Buil's magnet application.



Enter the two right-polarization images. If you are unsure whether they are right or left polarization, this is not critical; it affects the display values.

Enter the two left-polarization images.

Click the vertical-band correction option.

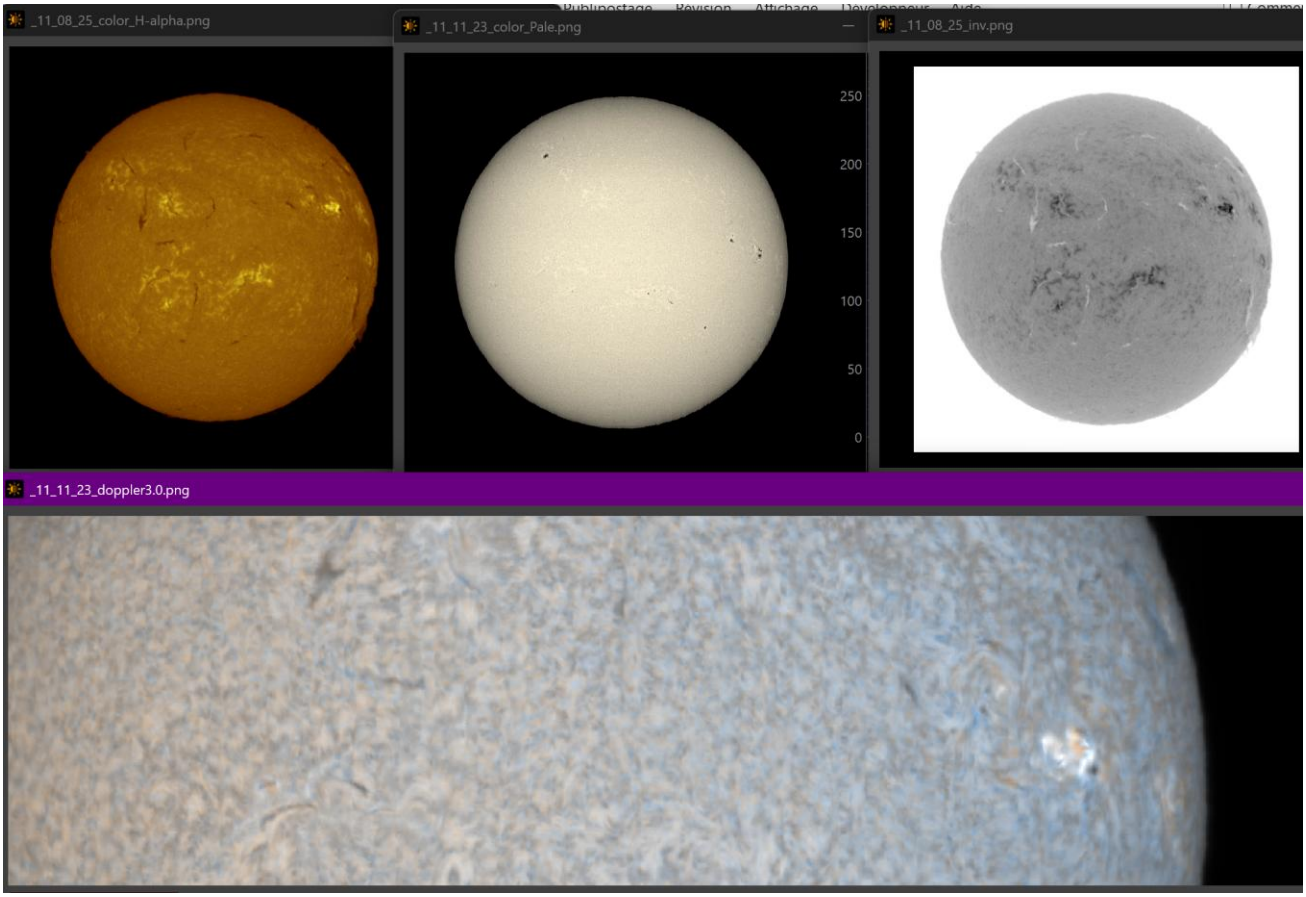
Click Go! and view your result in the mag-s.png file.

You can refine your image in the Processing tab and correct vertical transversallium bands.

To improve image quality, it is common to make several consecutive scans for each polarization. Simply select the files and let INTI process them in batch mode. INTI Partner will handle the summation in the magnet application.

Image viewing

The application can now open PNG image files for quick viewing from the General tab. To do this, change the file type selection using the “Open” button and choose, for example, PNG or FITS files.



Details of the inti_ini.yaml file

The INTI_ini.yaml file contains INTI configuration settings and preferences, except for interface settings.

At the end of a session, the settings are saved for the next session.

You can edit the YAML file in a text editor.

If you delete the file, or when INTI is launched for the first time, the application automatically recreates it.

- Directory: dernier répertoire des observations
- Dec doppler: décalage Doppler +/- en pixels
- Dec cont: décalage pour l'image de continuum
- poly_slit_a, poly_slit_b, poly_slit_c: cooefficients du polynôme
- ang_tilt: angle de tilt
- ratio_sysx: ratio géométrique pour la circularisation du disque
- free_autopoly: force le calcul automatique du polynôme sur la raie la plus sombre du spectre
- poly_free_a, poly_free_b,poly_free_c: coefficient du polynôme des modes spéciaux magnet et libre
- pos_free_blue: value in pixel of the position where to compute the blue part of the continuum in free lien mode
- pos_free_red: value in pixel of the position to compute the red part of the continuum in free line mode
- free_shift: shift to position the extraction at a different place than the polynome
- force_free_magn: force tilt and ratio de mise à l'échelle avec les valeurs entrées manuellement
- win_posx : obsolete
- win_posy: obsolete
- screen_scale: obsolete
- observer: alias de l'observateur pour entête fits
- site_long, site_ lat: site longitude et latitude
- instru: nom de l'instrument
- angle P: angle P utilisé – obsolete car mis à jour
- contact: adresse email
- wavelength: longueur d'onde du dernier scan
- wave_label: etiquette de longueur d'onde du dernier scan
- inversion NS: inversion Nord - Sud
- inversion EW: inversion Est - Ouest
- autocrop: active le mode autocrop
- ext20pct : extension de 20% en horizontal du mode crop auto
- trame couronne1 : index trame1
- trame couronne2 : index trame2
- pos fente min : pour récupérer des scans mal centrés - not for use
- pos fente max: pour récupérer des scans mal centrés - not for use
- zeeman_shift: decalage zeemanreduction_bruit: mode reduction de bruit
- zee_autopoly: force le calcul automatique du polynôme sur la raie la plus sombre du spectre
- grid disk: génère et enregistre image de la grille de StonyHurst
- lang: language d'INTI

Limitations

INTI's automated processing relies on a number of assumptions. For example, if the scan does not start before the solar limb appears and end after it disappears, INTI will be unable to determine the acquisition geometry. If the exposure is too low, insufficient contrast may also interfere with edge detection.

The only supported video format is SER: 8-bit or 16-bit, monochrome.

INTI is no longer compatible with Windows 7 because Windows 7 does not support Python versions later than 3.9.

INTI has been extensively tested by Christian Buil and myself on many different configurations. We are confident that it produces reliable results. Nevertheless, please contact us if you have any questions, problems, or suggestions.

References

INTI is an original work and an integral part of the Sol'Ex project, published in February 2021. If you use any of its components or are inspired by how it works, please acknowledge it as follows: Sol'Ex – INTI by Christian Buil & Valerie Desnoux.

Contributions

SER file reading is based on a Python library developed by Jean-Baptiste Butet, with SER timestamp decoding by Matt Considine.

The ellipse-fitting module is a Python module by Ben Hammel and Nick Sullivan-Molina (2020, March 21). bdhammel/least-squares-ellipse-fitting: v2.0.0 (Version v2.0.0). Zenodo. http://doi.org/10.5281/zenodo.3723294

The Python source code is available at https://github.com/Vdesnoux/inti**https://github.com/Vdesnoux/inti**

Appendix

Running an unsigned application on macOS

Place the file in a folder of your choice.

Before unzipping, open the macOS Terminal and enter the following command: sudo xattr -cr /pathtofile/specinti_mac.zip

Replace /pathtofile/ with the path to your ZIP file.

If the file has already been unzipped, specify the path to the folder instead.

Because sudo is a superuser command, the Terminal will ask for your password.

Type your Mac account password and press Enter. Nothing appears on the screen while you type the password. If the command succeeds, the Terminal displays no error message.

You can then unzip the file or open the folder directly. The first launch takes a long time and may take up to two minutes, so please be patient. Subsequent launches will be much faster.